Sebastian Capellan

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EDUCATION

New York University New York, NY
B.A. Data Science & Computer Science, GPA: 3.6/4.0 Expected May 2028
Relevant Coursework: Databases, Operating Systems, Computer Architecture, Data Structures, Algorithms, Linear Algebra
Activities: President & Founder of ColorStack @ NYU, Management Leadership for Tomorrow Fellow (MLT, SWE Track)

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL, Typescript
Technologies: DuckDB, DynamoDB, MySQL, Linux, Docker, AWS, Terraform, Postman, Grafana, Git, GitHub
Libraries & Frameworks: Flask, FastAPI, NumPy, Pandas, Numba, Pytest

WORK EXPERIENCE

Duolingo Pittsburgh, PA
Software Engineer Intern June 2026 - August 2026

- Built a social culture learning game with AI graded free text answers to location based prompts, implementing Flask API routes for submission delivery, handling, and scoring.
- Provisioned a serverless data architecture via DynamoDB and AWS S3 buckets for game data storage, using Terraform (HCL) for infrastructure and Jenkins for CI/CD build testing.
- Designed and implemented native iOS UI and gameplay logic using Swift in close collaboration with 3 other engineers.

University of California, San Francisco Remote
Research Assistant October 2025 - April 2026

- Accelerated a polymer research dashboard by 70%, replacing request-time file parsing with a singleton SQLite database.
- Engineered a universal Python ETL pipeline that auto-infers schemas from heterogeneous JSON sources, normalizes nested objects and arrays into relational SQLite tables, and validates row counts post-migration across 10+ tables.

PROJECTS

Geospatial Housing Price Heatmap | Python, SQL, DuckDB, Typescript

- Shipped "HexHousing," an interactive housing price heatmap visualizing prices as color-coded H3 hexagons on a map; built with MapLibre GL and deck.gl for clean UX and data visualization.
- Analyzed live service metrics reaching users across 10+ countries with 60k monthly requests and 10k page visits, determining an estimated 65% of traffic to be automated, informing scaling and rate limiting decisions.
- Architected an autonomous ETL pipeline that formats and ingests 26k+ US housing prices from Zillow and Census data into a DuckDB centric warehouse, enabling live parameterized user queries and filtering via FastAPI.

Quantitative Backtesting Engine | Python, SQL, DuckDB, Pandas, Pytest

- Engineered an asynchronous, ETL-pipelined, event-driven backtester using DuckDB and Alpaca Markets, scaling analytical workloads to 10M OHLCV rows at 1.5M+ rows/sec through columnar storage and vectorized execution.
- Engineered vectorized Bollinger Bands + RSI signal pipelines, reliably processing 129 securities with 0 failed backtests against SPY and buy-and-hold benchmarks, outperforming 42% of measured security buy & hold benchmarks.
- Automated live paper order execution via Alpaca API with position guards and buying-power-based position sizing.

Multithreaded Key-Value Server | C, POSIX, Linux

- Built a high-performance multithreaded datastore achieving ~10K ops/sec for reads, ~6K ops/sec for writes, and ~5K ops/sec for mixed workloads with <200 µs latency.
- Engineered concurrent client handling and local networking using Unix domain sockets, POSIX pthreads, and file locking, ensuring thread safe, low latency, and multi-client access to persistent storage.